

WENATCHEE PANGBORN, WA, USA

WMO: 727825

Lat: 47.398N

Lon: 120.201W

Elev: 375

StdP: 96.91

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94239

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.3	-9.6	-16.9	0.9	-10.1	-14.1	1.2	-7.6	9.6	4.4	8.1	4.2	1.2	290	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	35.7	18.1	33.8	17.7	31.9	17.1	19.5	32.8	18.6	31.3	17.8	30.0	3.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	10.6	24.2	13.1	9.8	23.8	12.3	9.3	23.4	56.8	32.8	53.8	31.2	51.1	30.0	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.1	DB	-14.7	38.6	3.8	1.8	-17.4	39.9	-19.6	40.9	-21.7	41.9	-24.5	43.2	(4)
(5)			WB	-15.2	20.6	3.6	1.0	-17.9	21.4	-20.0	21.9	-22.0	22.5	-24.6	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-1.4	1.4	6.4	10.8	15.7	19.4	24.2	23.5	18.4	10.6	3.5	-1.4	(6)	
	DBStd	9.72	4.17	4.07	3.58	3.38	4.01	3.84	3.58	3.35	3.68	3.68	4.06	4.04	(7)	
	HDD10.0	1329	353	239	119	29	3	0	0	0	1	34	197	354	(8)	
	HDD18.3	3161	611	473	371	227	102	31	2	3	44	239	446	613	(9)	
	CDD10.0	1692	0	0	7	53	181	282	441	420	252	54	2	0	(10)	
	CDD18.3	479	0	0	0	1	22	62	184	164	46	1	0	0	(11)	
	CDH23.3	5183	0	0	0	18	254	669	2106	1696	431	9	0	0	(12)	
	CDH26.7	2319	0	0	0	2	77	266	1057	777	138	2	0	0	(13)	
(14) Wind	WSAvg	3.1	1.5	1.9	2.9	3.9	4.1	4.5	4.4	4.0	3.3	2.6	1.9	1.5	(14)	
(15) Precipitation	PrecAvg	210	28	21	16	16	15	15	7	9	8	11	29	34	(15)	
	PrecMax	363	62	58	59	85	51	48	29	51	59	44	91	70	(16)	
	PrecMin	120	0	0	1	0	0	0	0	0	0	0	0	2	(17)	
	PrecStd	50	15	15	13	16	15	13	8	12	13	11	18	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	13.4	19.6	25.9	31.6	36.4	38.5	37.1	33.3	24.2	14.9	9.2	(19)	
		MCWB	5.2	7.4	10.6	13.6	15.9	18.4	18.8	19.1	17.9	14.1	9.4	4.9	(20)	
	2%	DB	6.9	10.3	16.6	22.4	28.9	32.4	36.3	35.0	30.2	21.0	12.4	6.9	(21)	
		MCWB	3.6	5.6	8.8	12.0	15.2	16.7	18.2	18.1	16.6	12.6	8.1	3.8	(22)	
	5%	DB	4.8	8.5	14.3	19.8	26.2	29.6	34.4	33.0	27.9	18.7	10.6	4.8	(23)	
		MCWB	2.4	4.5	7.4	10.5	14.1	15.7	17.7	17.9	15.9	11.2	7.1	2.5	(24)	
	10%	DB	2.9	6.8	12.5	17.5	23.6	27.2	32.4	31.1	25.8	16.9	9.0	3.1	(25)	
		MCWB	1.5	3.4	6.5	9.2	13.0	15.2	17.3	17.1	14.8	10.3	6.2	1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	8.0	11.2	14.6	17.3	19.3	20.8	20.6	18.9	15.1	10.6	5.6	(27)	
		MCDB	9.0	12.2	18.2	24.2	28.9	33.3	34.6	33.7	30.3	22.2	13.6	8.2	(28)	
	2%	WB	3.9	6.3	9.3	12.6	15.8	17.7	19.6	19.5	17.6	13.3	9.1	4.1	(29)	
		MCDB	6.4	9.7	15.7	21.1	26.8	30.2	33.2	32.4	28.6	19.0	11.3	6.1	(30)	
	5%	WB	2.7	4.9	8.0	11.1	14.7	16.6	18.7	18.5	16.4	12.0	7.8	2.8	(31)	
		MCDB	4.4	7.9	13.3	19.1	24.7	28.3	31.9	30.8	26.5	17.6	10.1	4.4	(32)	
	10%	WB	1.7	3.8	6.9	9.8	13.6	15.5	17.9	17.7	15.3	10.8	6.5	1.8	(33)	
		MCDB	2.8	6.3	11.7	16.8	22.3	26.0	30.5	29.5	24.3	16.0	8.7	3.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	7.1	9.6	11.0	11.9	12.2	13.7	13.3	12.5	10.0	6.5	4.4	(35)	
		MCDBR	7.7	9.6	12.2	14.1	15.5	15.2	15.8	15.4	15.1	12.2	8.6	7.3	(36)	
	5% WB	MCWBR	5.3	5.9	6.6	6.8	6.2	5.4	5.0	5.3	6.3	6.2	5.3	5.2	(37)	
		MCWBR	6.8	8.9	11.4	13.6	14.1	14.2	14.9	14.7	14.5	11.2	7.4	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.285	0.289	0.302	0.322	0.336	0.341	0.340	0.335	0.328	0.318	0.297	0.283	(40)	
		taud	2.430	2.504	2.530	2.461	2.455	2.463	2.474	2.485	2.496	2.515	2.521	2.460	(41)	
	Edn at Noon	Edn at Noon	808	891	929	935	928	921	917	909	886	841	791	767	(42)	
		Edh at Noon	66	76	87	103	107	107	105	99	89	75	61	58	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.10	2.08	3.21	4.76	5.86	6.55	7.27	6.12	4.44	2.64	1.33	0.88	(44)	
	RadStd	RadStd	0.14	0.26	0.29	0.36	0.42	0.44	0.39	0.36	0.30	0.27	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	-0.40	N/A	N/A	N/A	N/A	N/A	
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+38	

Nomenclature: See separate page